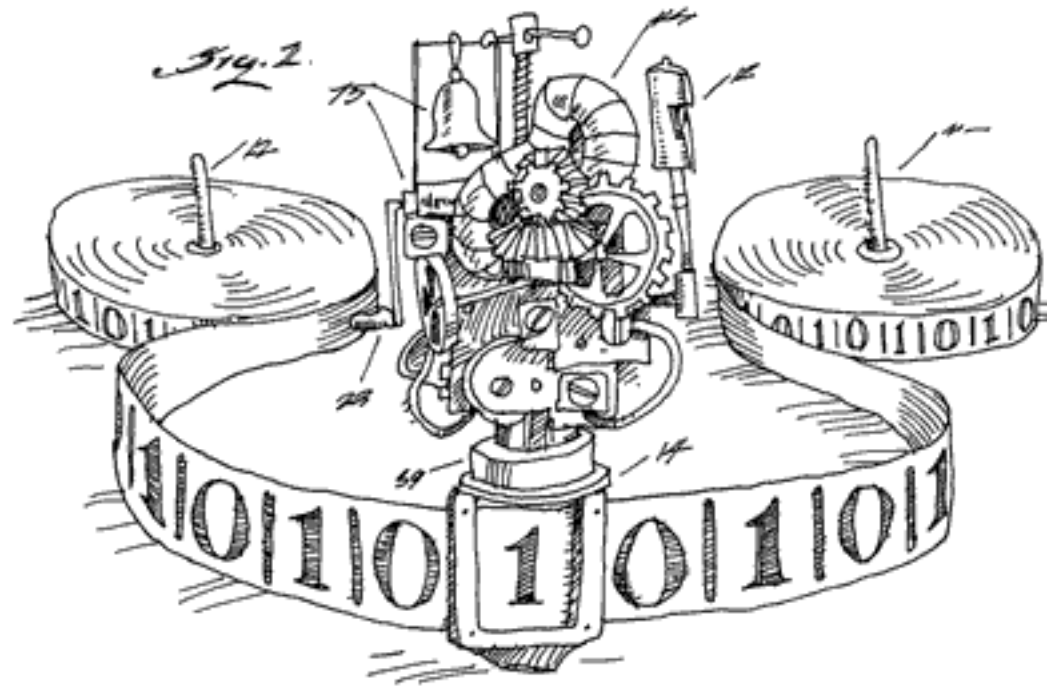


# CSC4010: Introduction to Theoretical Computer Science



## Lecture 9

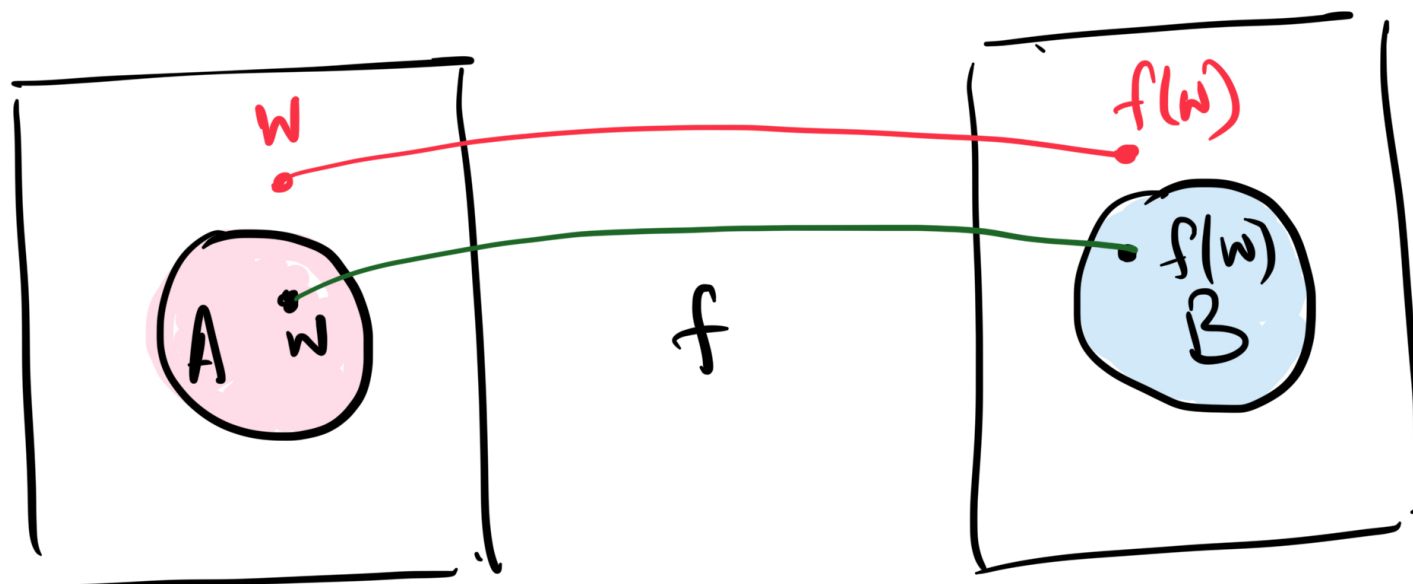
Adapted from Sipser's MIT OCW Course on TOC

**Recap**



# Mapping Reducibility

- **Definition:** A function  $f : \Sigma^* \rightarrow \Sigma^*$  is said to be **computable** if there exists a Turing Machine  $F$ , which on input  $w \in \Sigma^*$  terminates with just  $f(w)$  written on its tape, for any  $w$ .
- **Definition:** A language  $A$  is **mapping reducible** to a language  $B$  (denoted  $A \leq_m B$ ) if and only if there exists a computable function  $f : \Sigma^* \rightarrow \Sigma^*$  such that for every  $w \in \Sigma^*$ ,  $w \in A \iff f(w) \in B$ .



$f$  is called a mapping reduction from  $A$  to  $B$ .

$$A_{TM} \leq_m \overline{\text{EMPTY}}_{TM}$$

**Solution:** Given input  $\langle M, w \rangle$  for  $A_{TM}$ ,  $f(\langle M, w \rangle) = \langle M_w \rangle$  defined as follows:

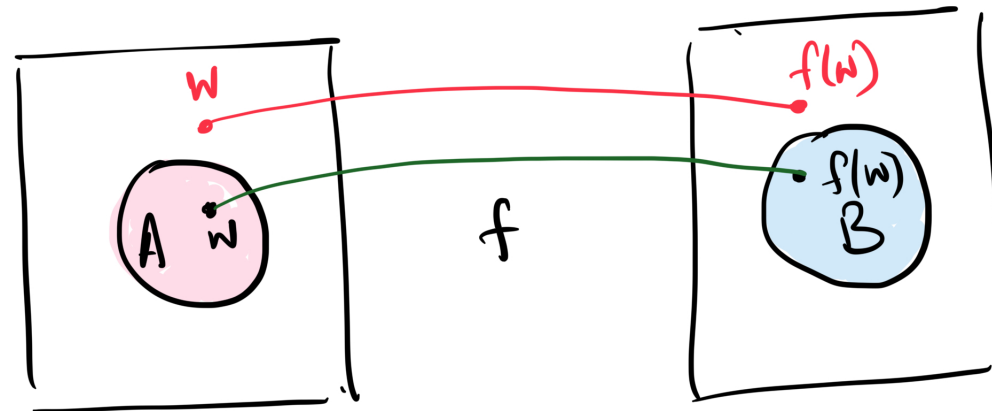
$M_w =$  "On input  $x \in \{0,1\}^*$

1. If  $x \neq w$ , then reject.
2. Simulate  $M$  on  $w$ .

**Claim:**  $f(\langle M, w \rangle) \in \overline{\text{EMPTY}}_{TM} \iff \langle M, w \rangle \in A_{TM}.$

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

# Mapping Reducibility: Properties



**Theorem:** If  $A \leq_m B$  and  $B$  is decidable (recognizable), then  $A$  is decidable (respectively, recognizable).

**Corollary:** If  $A \leq_m B$  and  $A$  is undecidable (unrecognizable), then  $B$  is undecidable (respectively, unrecognizable).

- **A recipe for undecidability/unrecognizability:** To show  $B$  is undecidable (unrecognizable), find undecidable (respectively, unrecognizable)  $A$  and a mapping reduction from  $A$  to  $B$ .

**Theorem:** If  $A \leq_m B$  then  $\bar{A} \leq_m \bar{B}$ .

# Using Mapping Reduction for Unrecognizability

# EMPTY<sub>TM</sub> is unrecognizable

- Let  $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$ .

**Theorem:** EMPTY<sub>TM</sub> is unrecognizable.

- Proof:** Proof by contradiction. We will show  $\bar{A}_{TM} \leq_m \text{EMPTY}_{TM}$ .

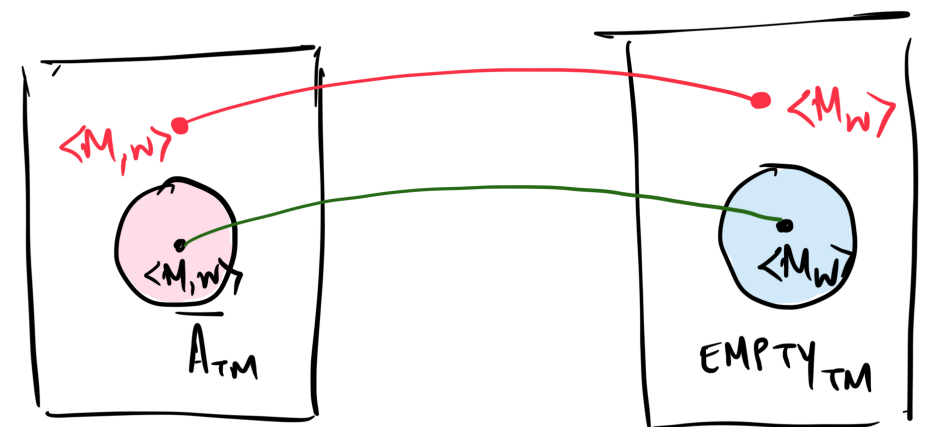
**Reduction Function:**  $f(\langle M, w \rangle) = \langle M_w \rangle$

$M_w =$  "On input  $x \in \{0,1\}^*$

1. If  $x \neq w$ , then reject.
2. Simulate  $M$  on  $w$ .

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

$$\begin{aligned} f(\langle M, w \rangle) \in \text{EMPTY}_{TM} &\iff \langle M_w \rangle \in \text{EMPTY}_{TM} \\ &\iff L(M_w) = \emptyset \\ &\iff M \text{ does not accept } w \\ &\iff \langle M, w \rangle \in \bar{A}_{TM} \end{aligned}$$



# More Examples

# $EQ_{TM}$ and $\overline{EQ}_{TM}$ are unrecognizable

- Let  $EQ_{TM} = \{ \langle M_1, M_2 \rangle \mid L(M_1) = L(M_2) \}$ .

**Theorem:**  $EQ_{TM}$  is unrecognizable.

- Proof:** Proof by contradiction. We will show  $\bar{A}_{TM} \leq_m EQ_{TM}$ .

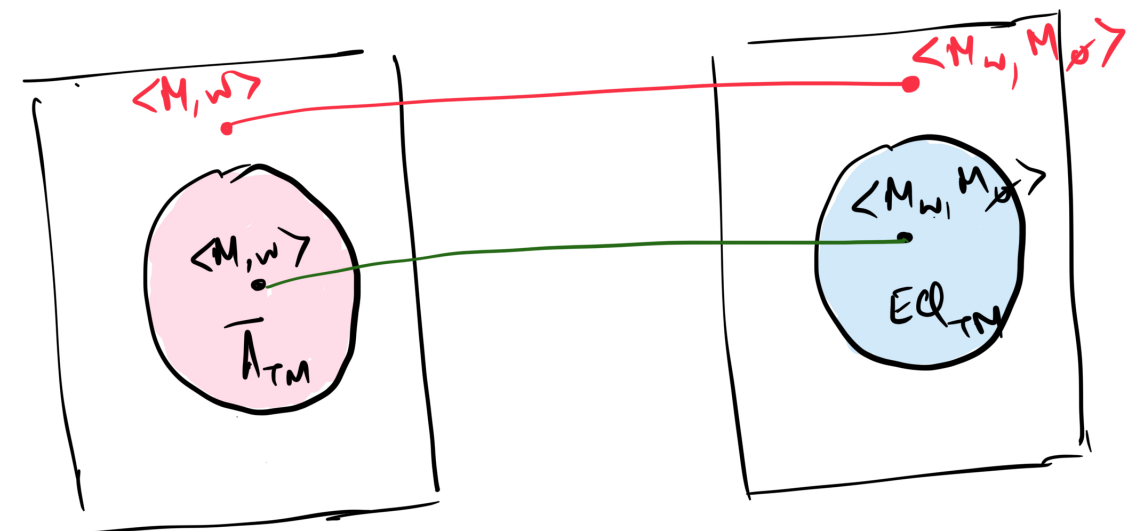
**Reduction Function:**  $f(\langle M, w \rangle) = \langle T_w, M_\emptyset \rangle$

$T_w =$  "On input  $x \in \{0,1\}^*$

1. Ignore  $x$ .
2. Simulate  $M$  on  $w$ .

$$L(T_w) = \begin{cases} \{0,1\}^* & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ rejects } w \end{cases}$$

$$\begin{aligned} \langle T_w, M_\emptyset \rangle \in EQ_{TM} &\iff L(T_w) = L(M_\emptyset) \\ &\iff L(T_w) = \emptyset \\ &\iff M \text{ does not accept } w \\ &\iff \langle M, w \rangle \in \bar{A}_{TM} \end{aligned}$$



# $EQ_{TM}$ and $\overline{EQ}_{TM}$ are unrecognizable

- Let  $\overline{EQ}_{TM} = \{ \langle M_1, M_2 \rangle \mid L(M_1) \neq L(M_2) \}$ .

**Theorem:**  $\overline{EQ}_{TM}$  is unrecognizable.

- Proof:** Proof by contradiction. We will show  $\overline{A}_{TM} \leq_m \overline{EQ}_{TM}$ .

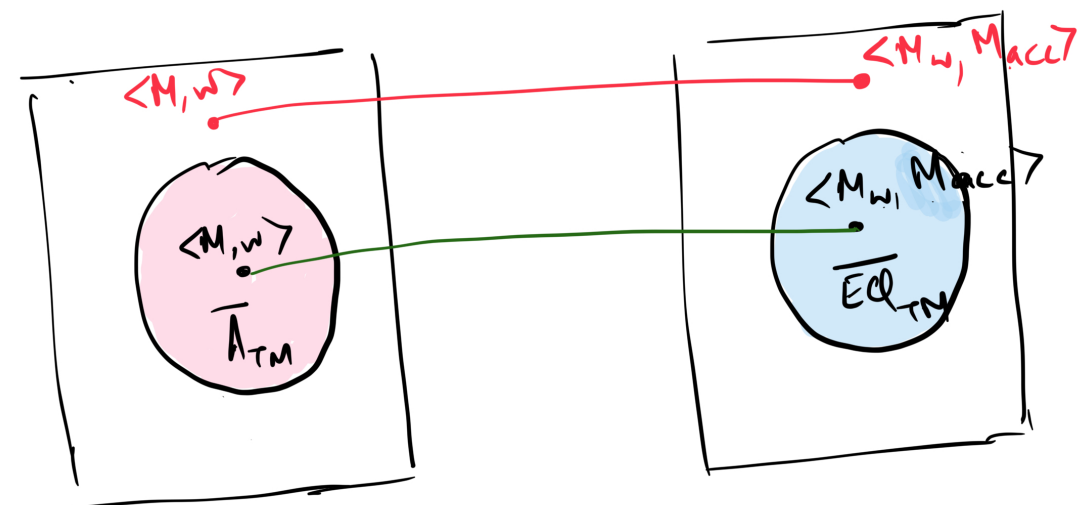
**Reduction Function:**  $f(\langle M, w \rangle) = \langle T_w, M_{acc} \rangle$

$T_w =$  "On input  $x \in \{0,1\}^*$

1. Ignore  $x$ .
2. Simulate  $M$  on  $w$ .

$$L(T_w) = \begin{cases} \{0,1\}^* & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ rejects } w \end{cases}$$

$$\begin{aligned} f(\langle T_w, M_{acc} \rangle) \in \overline{EQ}_{TM} &\iff L(T_w) \neq L(M_{acc}) \\ &\iff L(T_w) \neq \{0,1\}^* \\ &\iff M \text{ does not accept } w \\ &\iff \langle M, w \rangle \in \overline{A}_{TM} \end{aligned}$$



# Reducibility Terminology

Why do we use the term “reduce”?

- When we reduce  $A$  to  $B$ , we show how to solve  $A$  by using  $B$ . We conclude that  $A$  is no harder than  $B$ . (This suggests the  $\leq_m$  notation).
- Possibility 1: We bring  $A$ 's difficulty down to  $B$ 's difficulty.
- Possibility 2: We bring  $B$ 's difficulty up to  $A$ 's difficulty.

# Summary

- Introduced the mapping reducibility to show undecidability/unrecognizability.
- Showed  $\text{EMPTY}_{TM}$  is undecidable and unrecognizable.
- Showed  $\overline{\text{EQ}}_{TM}$  and  $\text{EQ}_{TM}$  are unrecognizable.